Inheritance

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1 Intro to inheritance

1.1 Intro to inheritance: Shapes

You want to build a software drawing various **2D Shapes**: Rectangle, Square, Polygon, Circle, Ellipse, etc.

- You can decide to build each class representing each particular shape. But you recognize "Wait a minute":
  - A `Square` is a `Rectangle`
  - A `Rectangle` is a `Polygon`
  - A `Circle` is a `ELlipse`
  - All of them is a `Shape`
- Furthore, we notice

- A `Polygon` has a list of `Point2D`
- A `Ellipse` has a center of `Point2D` and has two two axis sizes `a` and `b` of type `double`.
- A `Circle` is similiar to `Ellipse` but `a == b`.
- A `Rectangle` has a has a corner of `Point2D` and two sizes `a` and `b` of type `double`
- A `Square` is similar to a `Rectangle` but `a == b`.

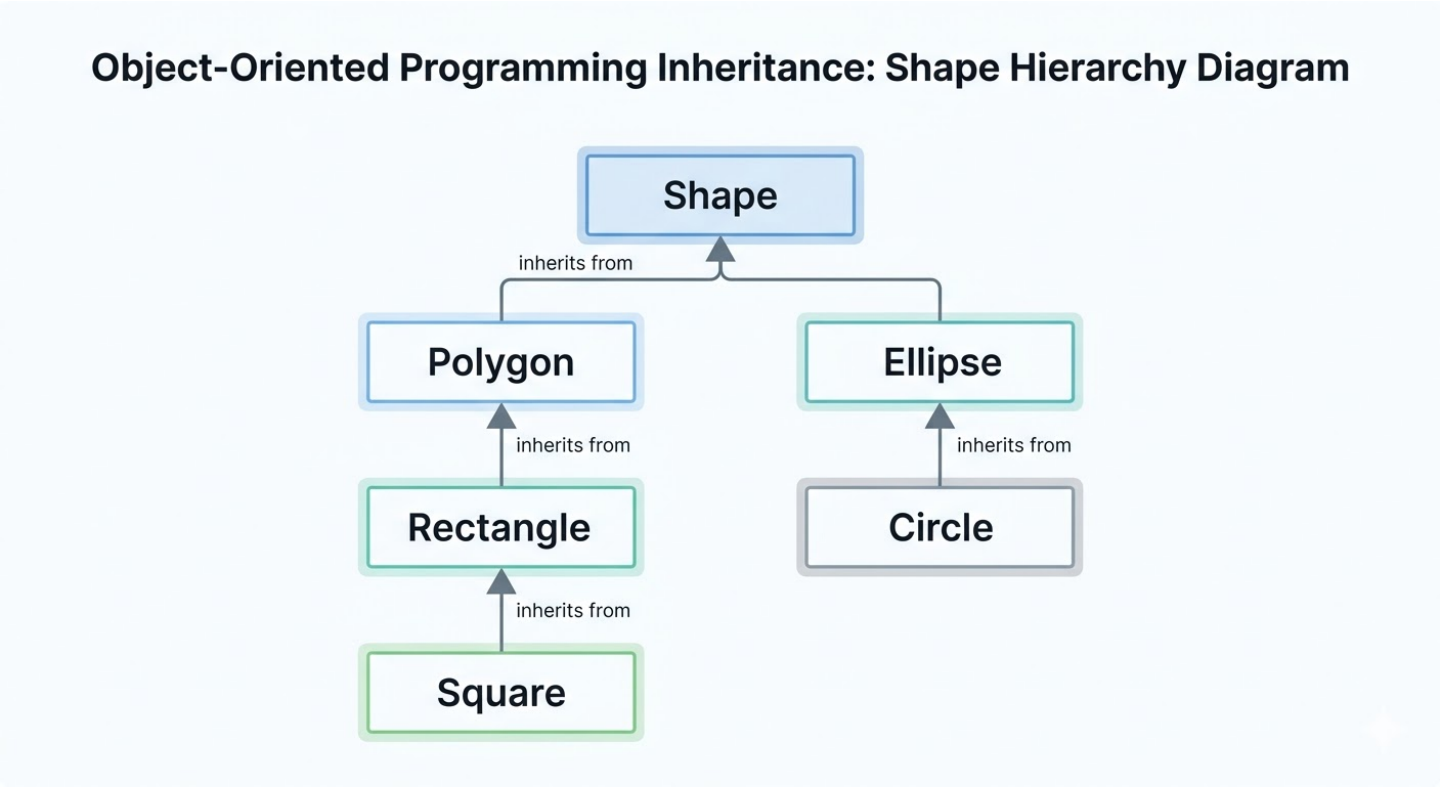
1.2 Intro to inheritance: Shapes

- You notice that
  - Drawing a square should be a consequence of drawing a rectangle with the same size.
  - Drawing a rectangle should be a consequence of drawing a "special" polygon of 4 sides.
  - Drawing a circle should be a consequence of drawing an ellipse

So why do we bother to build different classes separately!

1.3 Hierarchies

A hierarchy is a diagram showing how various objects are related.



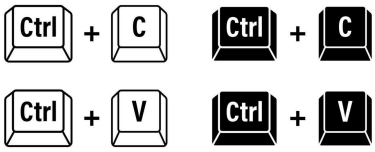
1.4 Intro to inheritance: University People

If the above example cannot convince you enough, look at our University hierarchy.

A very small description about roles in our Uni

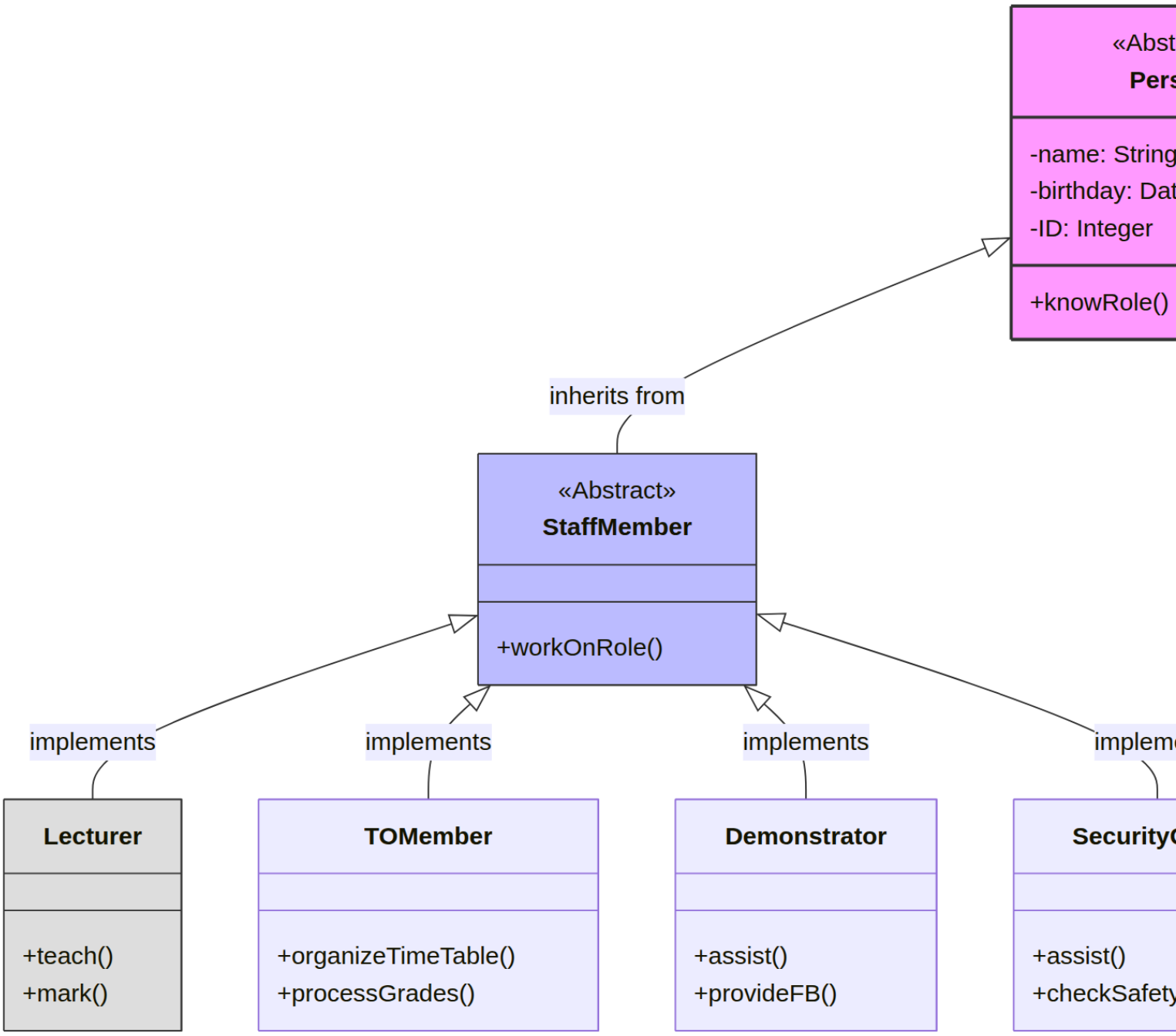
- Every member of staffs has a `name`, a `birthday`, an `ID`
- Every student has a `name`, a `birthday`, an `ID`
- There are different types of staff members: `Lecturer`, `TOMember`, `Demonstrator`, `SecurityGuard`
- There are different types of students: `UGStudent` (undergraduate), `PGTStudent` (Post-graduate Taught) and `PGRStudent` (Post-graduate Research)
- Every person in the Uni has their role and knows their role, work on their role.

If you are about to create all the classes `Lecturer`, `TOMember`, `Demonstrator`, `SecurityGuard`, `UGStudent`, `PGTStudent`, `PGRStudent`, you see that you must [*plug figure here*] a lot for common data members and member functions.



This violates the principle designing rule: **Do not repeat yourself**

1.5 Intro to inheritance: How about the following design hierarchy



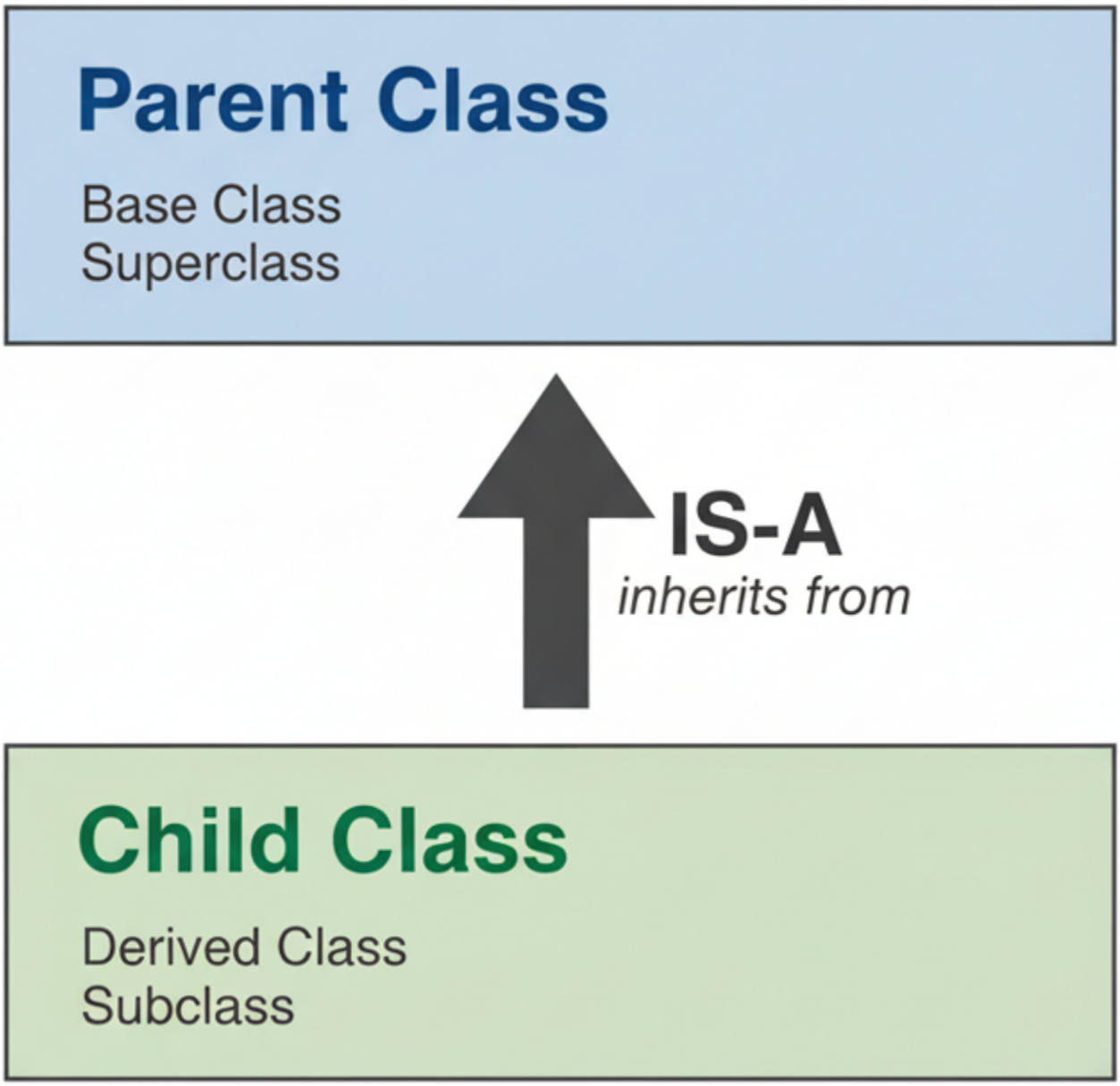
2 Basic inheritance in C++

2.1 Terminilogies

2.2 Terminilogy

In an inheritance (is-a) relationship

- the class being inherited from: **parent class**, **base class**, or **superclass**
- the class doing the inheriting: **derived class**, **derived class**, or **subclass**



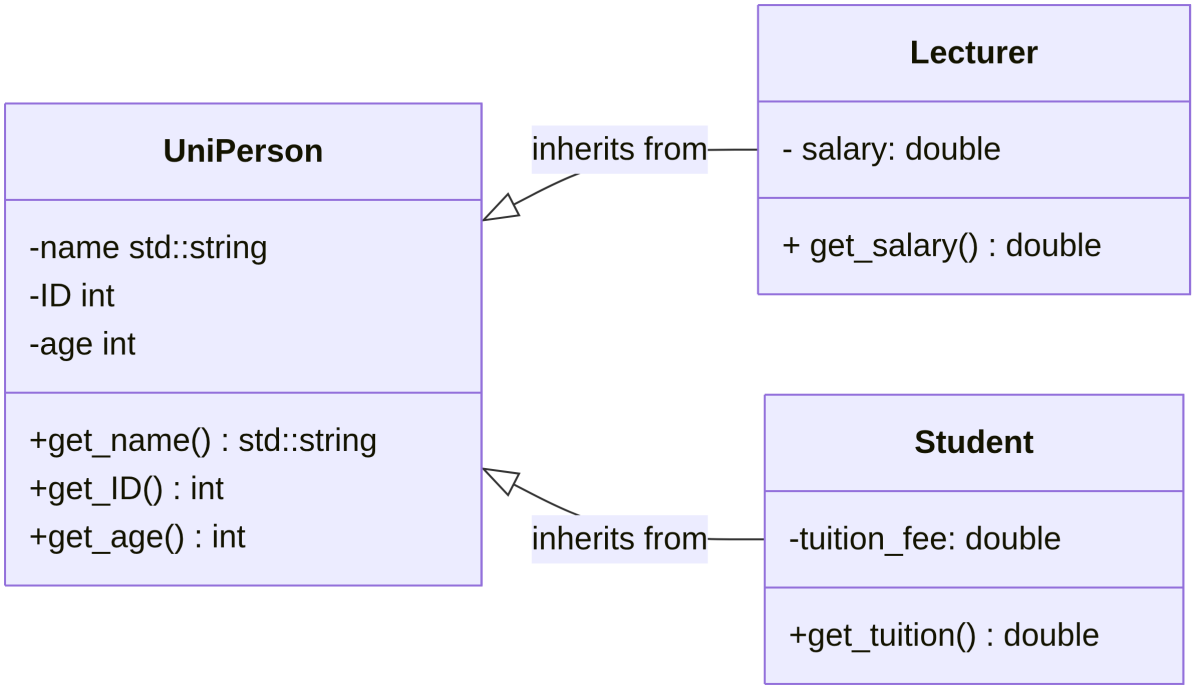
2.3 Basic inheritance in C++: Mini example

We will adopt a smaller design structure:

- Every `Lecturer` is a a `UniPerson`
- Every `Student` is a a `UniPerson`
- Every `Lecturer` and `Student` has `name` and `ID`.

We shall discuss - `work_on_role()` later when we study **Polymorphism** and **Virtual Function**. - other member functions in `Lecturer` and `Student` when we study [*adding more functionality*] to derived class.

2.4 Basic inheritance in C++: Mini-diagram for code demonstration



2.5 Implementation of Inheritance: class `UniPerson`

Let us look at the class `UniPerson`

2.6 Implementation of Inheritance: class `Lecturer` without inheritance

Question Can you tell me what problem we have?

2.7 Implementation: class `Lecturer` inherited from class `UniPerson`

2.8 Impelemntation: Put two classes together into a `main` program

Notice that we can access `age`, `id` members although we did not define them explicitly in class `Lecturer`. *Let us run the code.*

2.9 Implementation: class `Student`

It is now easy to write class `Student` derived from `UniPerson`

2.10 Implementation: Put three classes together into a `main` program

Listing 1 class `UniPerson`. Filename=`uniperson.cpp`

```cpp
#include <string>
#include <string_view>

class UniPerson
{
private:
    // This is shared by all Person objects
    static int nextID;
    std::string name{};
    int ID {};
    int age {};
public:
    UniPerson(std::string_view name = "", int age = 0)
        : name{ name }, age{ age }
    {
        // Assign the current counter and
        // then increment it for the next object
        ID = nextID++;
    }

    const std::string& get_name() const { return name; }
    int get_age() const { return age; }
    int get_id() const { return ID; }
};

int UniPerson::nextID{100000};
```

2.11 Inheritance chains

We can comeback to our bigger diagram

`UGStudent`, `PGTStudent` and `PGRStudent` are all derived from class `Student` and thus we have **inherintance chain**

Listing 2 class `Lecturer` without inheritance. Filename=`lecturer_noinheritance.cpp`

```cpp
#include <string>
#include <string_view>

class Lecturer  // non inheritance from UniPerson
{
    static int nextID;
    double salary;
    std::string name;
    int age;
    int ID;
public:


    Lecturer(std::string name, int age): name{name}, age {age}
    {
        // Assign the current counter and
        // then increment it for the next object
        ID = nextID++;
    };
    // we must copy + paste what we wrote for UniPerson

    int age() { return age; }
    const std::string& name() { return name; }
};

// you immediately see the problem here
// we just count the ID from all over again.
int Lecturer::nextID{10000};
```

Listing 3 class `Lecturer` inherited from class `UniPerson`. Filename=`lecturer.h`

```
#ifndef LECTURER_H
#define LECTURER_H

#include <string>
#include <string_view>
#include "uniperson.h"

class Lecturer : public UniPerson  // non inheritance from UniPerson
{
private:
    double salary;
    // We can remove a lot of code here!!!
public:
    Lecturer(std::string_view name = "", int age = 0, double salary = 42000):
        UniPerson{name, age}, // we borrowed the constructor from UniPerson above
        salary {salary}

    {
    };
    double get_salary() const { return salary; }
    // We can remove a lot of code for member functions here.
};
// We don't need this code anymore and it makes perfect sense.
// int Lecturer::nextID{10000};

#endif
```

We will write a class for `UGStudent`.

Instead of writing everything all the classes in one single giant filet, let us do this in multiple header files, which is the next topic.

3 Split the program into header files and implementation files

3.1 Goals of this example

We go through the steps that need to be done to build a program that needs implementation of many classes.

- Write header files (`filename.h` or `filename.hpp`) for each class
- Write implementation files (`filename.cpp`) for the corresponding class
- Compile the main program using the `.cpp` files.

We build four classes and one main program. Each class is done by one header file `.h` and one implementation file `.cpp`:

- class `UniPerson`
- class `Lecturer` inherits from `UniPerson`
- class `Student` inherits from `UniPerson`
- class `UGStudent` inherits from `Student`

3.2 class `UniPerson`: Header file

3.3 class `UniPerson`: Implementation file

3.4 class `Lecturer`: Header file

3.5 class `Lecturer`: Implementation file

3.6 class `Student`: Header file

3.7 class `Student`: Implementation file

3.8 class `UGStudent`: Header file

3.9 class `UGStudent`: Implementation file

4 Order of construction of derived classes

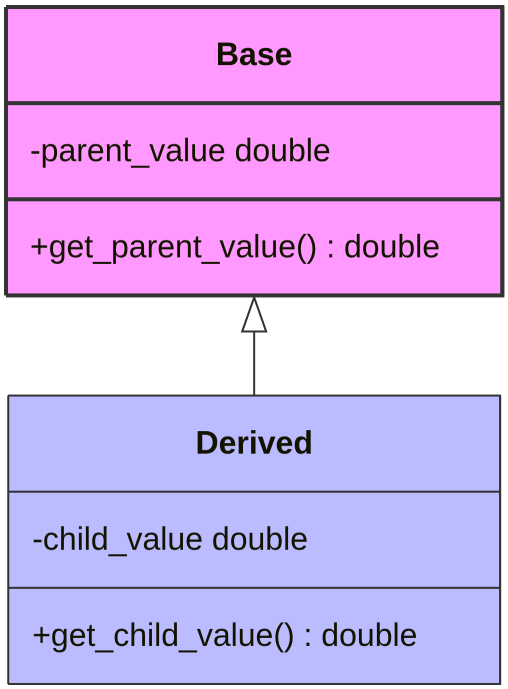
In this lesson, we take a closer look at the order of construction that happens when a derived class is instantiated.

4.1 Order of constructors

Let us look at this simple code involving two classes

4.2 Order of construction of derived classes: Diagram

The diagram of the above code snippet is easy:



The members of `Base` are **not** copied into `Derived`.

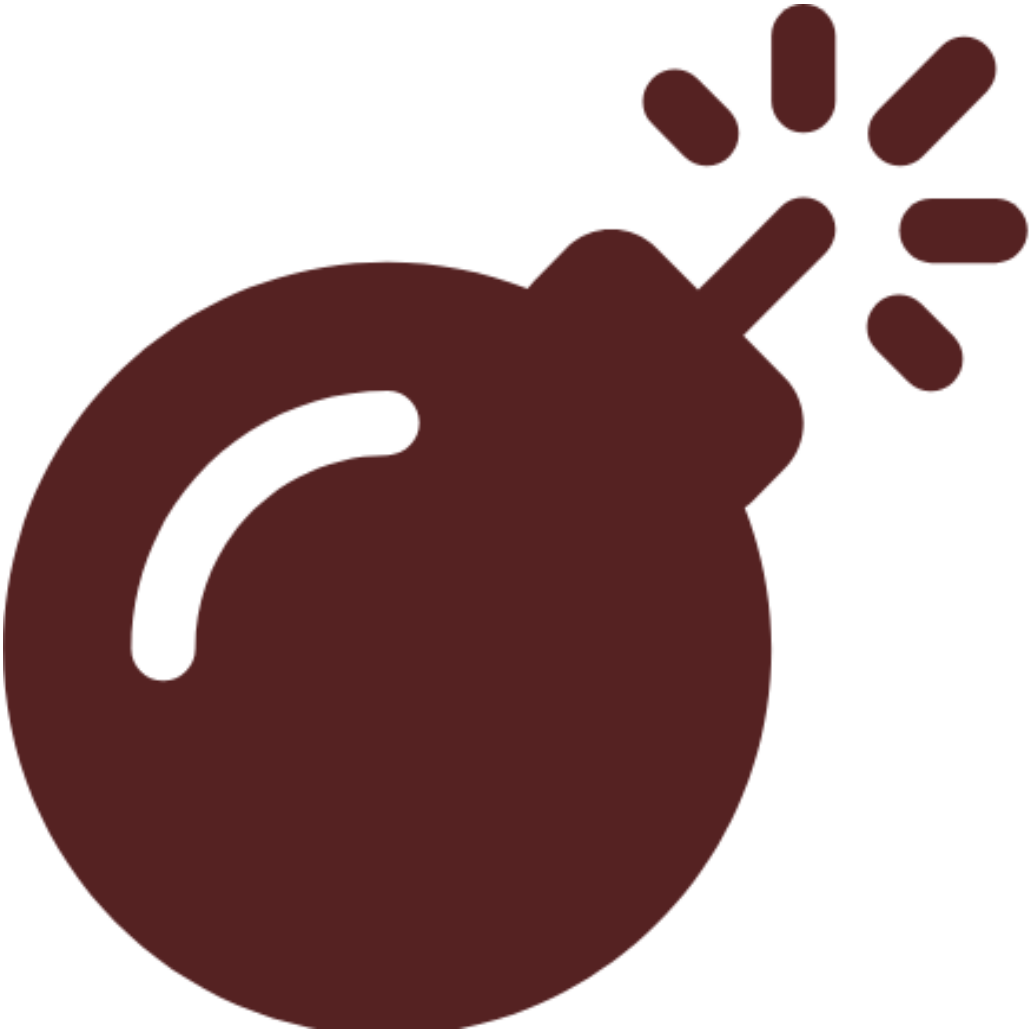
Instead, we can consider `Derived` and `Base` as a two-part class: one part `Derived` + one part `Base`

4.3 Order of construction of derived classes: A closer look

The constructor of `Base` class is executed before the constructor of `Derived` class is excuted when a `Derived` object is created.

4.4 Order of construction for inheritance chains

Let us look at this example



Question If I create an object of class `D`, what would you expect to be executed?

The above idea/knowledge applies naturally to **inheritance chains**:

- the great grandparent `A` constructor is executed, **then**
- the grandparent `B` constructed is executed, **then**
- the parent `C` constructor is executed, and **then**
- the derived `D` constructor is executed.

4.5 Order of construction for inheritance chains: Code demonstration

source-code

5 Constructors and initialization of derived classes

In Python, we can use `super()` to access the parent's data and member functions.

Thus, we can write `super().__init__(self,...)` to use the constructor of the parent class.

```
class UniPerson:

    nextID = 10000  # class variable

    def __init__(self, name="", age=18):
```

```python
        self.name, self.age = name, age
        self.id = UniPerson.nextID
        UniPerson.nextID += 1
class Lecturer(UniPerson):
    def __init__(self, name="", age=18):
        super().__init__(name, age)
        print(f"Lecturer {self.name} is recruited.")
```

This will be the topic of this lesson.

5.1 Revisit Python example

```python
class UniPerson:
    nextID = 10000  # class variable, functioning like static int nextID in C++
    def __init__(self, name="", age=18):
        self.name, self.age = name, age
        self.id = UniPerson.nextID; UniPerson.nextID += 1   # fit into the slide
class Lecturer(UniPerson):
    def __init__(self, name="", age=18):
        super().__init__(name, age)
        print(f"Lecturer {self.name} is recruited.")
    def __str__(self):      # print out the name of the lecturer
        return self.name

khiem = Lecturer("Khiem", 42)
print(f"ID assigned for {khiem}: {khiem.id}")
```

```
Lecturer Khiem is recruited.
ID assigned for Khiem: 10000
```

5.2 Initialization involving inheritance: A problem example

Look at this simple example

5.3 Initialization: Looking at `Base` class

```
Base parent{ 42 };
```

What happens when `Base` is instantiated:

- Memory for base class (`Base`) is set aside.
- Appropriate `Base` constructor is called.
- Member initializer list initializes variables
- Body of constructor executes.
- Control is returned to the caller.

5.4 Initialization: Looking at `Derived` class

```
Derived derived{ 42.24 };
```

What happens when `Derived` is instantiated:

- Memory for derived class (`Derived`) is set aside.
- Appropriate `Derived` constructor is called.
- *`Base` object is constructed first using the pappropriate `Base` constructor*. If no parent constructor is specified, the default constructor will be used.
- Member initializer list initializes variables.
- Body of the constructor executes.
- Control is returned to the caller.

Question

- How about the `id` member that a `Derived` object inherits from `Base`?
- How can we initialize it when we create an object of `Derived` class.

5.5 Initializing base class members: Approach 01

```
class Derived: public Base
{
public:
    double cost {};

    Derived(double value=0.0, int id=0)
        : value { value }, id { id }  // this won't work
    {}

    double get_cost() const { return cost; }
};
```

The value of a member variable can only be set in a member initializer list of a constructor belonging to the same class as the variable. This attempt won't work. See and run `initialization_of_derived_class_approach_01.cpp`

Question What else can we do?

5.6 Initializing base class members: Approach 02

```
class Derived: public Base
{
public:
    double cost {};

    Derived(double value=0.0, int id=0)
        : value { value }
    {   // this work but not work for a const or a reference
        this->id = id;
    }

    double get_cost() const { return cost; }
};
```

This approach works. See and run `initialization_of_derived_class_approach_02.cpp`. **But**

- **would not work if `id` were a const or a reference**. const values and references have to be initialized in the member initializer list of the constructor.
- **inefficient**. Member `id` gets assigned a value twice: once in the member initializer list of `Base` class constructor, and then again in the body of the `Derived` class.

5.7 Initializing base class members: Call parent class constructor

C++ allows us to explicitly choose which Base (Base) class constructor will be called! Simply add a call to the Base class constructor.

```
class Derived: public Base
{
public:
    double value {};

    Derived(double value=0.0, int id=0)
        : Base{ id } // call Base(int) constructor with value id!!!
        , value{ value }
    { } // constructor body

    double get_value() const { return value; }
};
```

This resolves the troubles we have above.

5.8 Initializing base class members: Put things together

Let us run the code snippet

5.9 Initializing base class members: `UniPerson` and `Student`

Homework

Think about rewriting `Student` so that we can call `UniPerson`'s constructor in the member initializer list of `Student`

5.10 Inheritance chains

Let us apply the concept we have learned to inheritance chain

Question What's the output on the console?

5.11 Destructors

When a derived class is destroyed, each destructor is called in the reverse order of construction.

In the above example, when `c` is destroyed, the `C` destructor is called first, then the `B` destructor, then the `A` destructor.

6 Inheritance and access specifiers

We have learned two access specifiers:

- private
- public

We have also seen the access specifier `protected`.

We will learn more about how access specifiers interact with inheritance in this lesson.

6.1 The `protected` access specifier

The access specifier `protected` is only useful in an inheritance context.

- The `protected` access specifier allows the class the member belongs to, friends, and derived classes to access the member.
- `protected` members are not accessible from outside the class.

6.2 Access specifiers in inheritance: Code illustration

6.3 When should we use the `protected` access specifier

Discuss briefly in the lecture. *Students need to read these points carefully again.*

- With a `protected` attribute in a base class, derived classes can access that member directly. If we later change anything about the `protected` attribute (type, value meaning, etc.), we'll probably **need to change both the base class and all of the derived classes**.
- Using the `protected` access specifier is most useful when
  - we (and our team) are going to be the ones deriving from our own classes
  - number of derived classes is reasonable
- Making members private is good for insulating the public or derived classes from implementation changes, and for ensuring invariants (fractions must have non-zero denominators) are maintained properly.
  - Think about the open-source code and you made member protected. Other developers can access the members and spoil them.
  - This implies our class need a larger public (or protected) interface to support all of the functions that public or derived classes need for operation (high cost to build and maintain)
- It's better to make your members private if we can, and **only use `protected` when derived classes are planned** and the cost to build and maintain an interface to those private members is too high.

6.4 When should we use the `protected` access specifier

> ❗ Important
>
> **Best practice**
> Favor private members over protected members.

6.5 Different kinds of inheritance, and their impact on access

There are three different ways for classes to inherit from other classes: `public`, `protected`, `private`

We have 9 combinations: 3 member access specifiers and 3 inheritance types.

```
// Inherit from Base publicly
class Pub: public Base {
};

// Inherit from Base protectedly
class Pro: protected Base {
};

// Inherit from Base privately
class Pri: private Base {
};

class Def: Base {// Defaults to private inheritance
};
```

6.6 Different kinds of inheritance, and their impact on access

Some remarks

- It is not that I cannot teach this component carefully in the live lecture.
- I think we will get confused and forget most of the explained cases right after leaving the classroom.
- The point is to make you aware of these concepts and give you an opportunity to learn them deeper.

Therefore, I write down some keypoints and skip the slides. *Students should read the materials later.*

Tip

Key rules to interpret different combinations

- A class can always access its own (non-inherited) members.
- The public accesses the members of a class based on the access specifiers of the class it is accessing.
- A derived class accesses inherited members based on the access specifier inherited from the parent class. This varies depending on the access specifier and type of inheritance used.

6.7 `public` inheritance

- `public` inheritance is by far the most commonly used type of inheritance.
- Very rarely will we see or use the other types of inheritance.

Access specifier in base class	Access specifier when inherited publicly
Public	Public
Protected	Protected
Private	Inaccessible

6.8 `protected` inheritance

- `protected` inheritance is the least common method of inheritance.
- It is almost never used, except in very particular cases.
- With `protected` inheritance, the `public` and `protected` members become `protected`, and `private` members stay inaccessible.

Access specifier in base class	Access specifier when inherited protectedly
Public	Protected
Protected	Protected
Private	Inaccessible

6.9 `private` inheritance

- With `private` inheritance, all members from the base class are inherited as `private`.
- `private` members are inaccessible, and `protected` and `public` members become `private`.

Access specifier in base class	Access specifier when inherited privately
Public	Private
Protected	Private
Private	Inaccessible

6.10 Access specifier in inheritance: Summary

Base Class Member	Public Inheritance	Protected Inheritance	Private Inheritance
Public	Public	Protected	Private
Protected	Protected	Protected	Private
Private	Inaccessible	Inaccessible	Inaccessible

7 Adding new functionality to a derived class

One of the biggest benefits of inheritance is the **ability to reuse written code**.

We can **add new functionality, modifying existing functionality, or hide functionality** if we don't want.

> We have learned these concepts from **Introductory Programming 2** and also in several last examples. No need to go into too much details.

7.1 Adding new functionality to a derived class: Code illustration

Let us run it and go through it slowly.

7.2 Adding new functionality: An example using `std::vector`

Let us run it and go through it slowly.

8 Calling inherited functions and overriding behavior

By default, derived classes inherit all of the behaviors defined in a base class.

We now examine in more detail how member functions are selected, as well as how we can leverage this to change behaviors in a derived class.

> **Note**
>
> - Most of us have learned all these concepts in **Introductory Programming 2** and **Mechanical Engineering Skills 3**.
> - It does not harm to quickly go through them again. Well, people can forget .

8.1 Calling a base class function

Since the derived class inherits members of its base class, it can use them (need access specifier `public` or `protected` in base class).

Let us run this code snippet.

8.2 Redefining behaviors

However, if we had defined `Derived::identify()` in the Derived class, it would have been used instead.

8.3 Adding to existing functionality

Sometimes we do not want to completely replace a base class function, but instead want to add additional functionality to it when called with a derived object.

9 Change an inherited member's access level and hide inherited functionality

Change an inherited member's access

C++ gives us the ability to change an inherited member's access specifier in the derived class.

Hide functionality

In a derived class, it is possible to hide functionality that exists in the base class, so that it can not be accessed through the derived class.

We can do both with `using` declaration.

We skip this component. *The students can learn it at home.*

9.1 Change an inherited member's access level: make `public`

C++ gives us the ability to change an inherited member's access specifier in the derived class, by a `using` declaration – Let us run this code snippet:

9.2 Change an inherited member's access level: hide functionality (make `private`)

In a derived class, it is **possible to hide functionality that exists in the base class**, so that it can not be accessed through the derived class – Let us play around this code snippet:

9.3 Change an inherited member's access level: Summary

With `using` declaration, we can change an inherited member's access level

- make the `member` in `Derived` class `public` by declaring `using Base::member;` below access specifier `public:`
- hide the `member` in `Based` class by declaring `using Base::membe;` below access specifier `private:`

9.4 Deleting functions in derived class

We can mark member functions as deleted in the derived class, ensuring they cannot be called at all though a derived object.

10 Multiple inheritance

Multiple inheritance enables a derived class to inherit members from more than one parent.

> **Note**
>
> - Again, I expect that most of us (students) have learned all these concepts in **Introductory Programming 2** and **Mechanical Engineering Skills 3**.
> - However, I can acknowledge that maybe many of you have not heard about this.

10.1 Multiple inheritance: Smartphone – A Concept hierarchy

Let us think of this example

- `Camera`: provides the ability to `take_photo()`
- `Phone`: provides the ability to `make_call()`
- `Smartphone`: inherits from both, allowing it to perform both actions.

10.2 Multiple inherintance: Smartphone – Code

Listing 4 A program using class `Lecturer` derived from the class `UniPerson` Filename=`main_uniperson_lecturer.cpp`

```cpp
#include <string>
#include <string_view>
#include <iostream>
class UniPerson
{
protected:  // Protected so that Lecturer can see these
    // This is shared by all Person objects
    static int nextID;
    std::string name{};
    int ID {};
    int age {};
public:
    UniPerson(std::string_view name = "", int age = 0)
        : name{ name }, age{ age }, ID {nextID++}
    {
    }

    const std::string& get_name() const { return name; }
    int get_age() const { return age; }
    int get_id() const { return ID; }

};

int UniPerson::nextID{100000};

class Lecturer : public UniPerson  // non inheritance from UniPerson
{
private:
    double salary;
    // We can remove a lot of code here!!!
public:


    Lecturer(std::string_view name = "", int age = 0, double salary = 42000):
        UniPerson{name, age}, salary {salary}
    {
        // we borrowed the constructor from UniPerson above
    };
    double get_salary() const { return salary; }
    // We can remove a lot of code for member functions here.
};

// We don't need this code anymore and it makes perfect sense.
// int Lecturer::nextID{10000};
int main()
{
    Lecturer khiem {"khiem", 42};
    std::cout << "Khiem's age is " << khiem.get_age() << '\n';
    std::cout << "Khiem's ID is " << khiem.get_id() << '\n';

    Lecturer scott {"Scott", 101}; // his hair tells me that
```

Listing 5 class `Student` inherited from the class `UniPerson` Filename=`student.h`

```
#ifndef STUDENT_H
#define STUDENT_H

#include <string>
#include <string_view>
#include "uniperson.h"

class Student : public UniPerson  // non inheritance from UniPerson
{
private:
    double tuition;
    // We can remove a lot of code here!!!

public:
    Student(std::string_view name = "", int age = 0, double tuition = 42000):
        UniPerson{name, age}, // we borrowed the constructor from UniPerson above
        tuition {tuition}
    {
    };

    double get_tuition() const { return tuition; }
    // We can remove a lot of code for member functions here.
};

#endif
```

Listing 6 A program using classes `Lecturer` and `Student` derived from the class `UniPerson` Filename=`main_uniperson_lecturer_student.cpp`

```cpp
#include <string>
#include <string_view>
#include <iostream>

class UniPerson
{

protected:  // Protected so that Lecturer can see these
    // This is shared by all Person objects

    static int nextID;
    std::string name{};

    int ID {};
    int age {};

public:
    UniPerson(std::string_view name = "", int age = 0)

        : name{ name }, age{ age }, ID {nextID++}
    {

    }


    const std::string& get_name() const { return name; }
    int get_age() const { return age; }

    int get_id() const { return ID; }


};


int UniPerson::nextID{100000};


class Lecturer : public UniPerson  // non inheritance from UniPerson
{

private:
    double salary;

    // We can remove a lot of code here!!!
public:



    Lecturer(std::string_view name = "", int age = 0, double salary = 42000):

        UniPerson{name, age}, salary {salary}
    {

        // we borrowed the constructor from UniPerson above
    };
```

Listing 7 Header file for class `UniPerson` Filename=`uni-program/uniperson.h`

```cpp
#ifndef UNIPERSON_H
#define UNIPERSON_H

#include <string>
#include <string_view>
class UniPerson
{
private:
    // This is shared by all Person objects
    static int nextID;
    std::string name{};
    int ID {};
    int age {};
public:
    // forward declare everything here
    UniPerson(std::string_view name = "", int age = 0);
    const std::string& get_name() const;
    int get_age() const;
    int get_id() const;
};

#endif
```

Listing 8 Implementation file for class `UniPerson` Filename=`uni-program/uniperson.cpp`

```cpp
#include "uniperson.h"

int UniPerson::nextID{100000}; // Static initialization here only!

UniPerson::UniPerson(std::string_view n, int a)
: name{n}, age{a}, ID{nextID++}
{}
const std::string& UniPerson::get_name() const {
    return name;
}
int UniPerson::get_age() const {
    return age;
}
int UniPerson::get_id() const {
    return ID;
}
```

Listing 9 Header file for class `Lecturer` Filename=`uni-program/lecturer.h`

```
#ifndef LECTURER_H
#define LECTURER_H

#include <string>
#include <string_view>
#include "uniperson.h"

class Lecturer : public UniPerson  // non inheritance from UniPerson
{
private:
    double salary;
    // We can remove a lot of code here!!!
public:
    Lecturer(std::string_view name = "", int age = 0, double salary = 42000);
    double get_salary() const;
};

#endif
```

Listing 10 Implementation file for class `Lecturer` Filename=`uni-program/lecturer.cpp`

```
#include "lecturer.h"

Lecturer::Lecturer(std::string_view name, int age, double salary)
    : UniPerson{name, age}, salary{salary} {}

double Lecturer::get_salary() const {
    return salary;
}
```

Listing 11 Header file for class `Student` Filename=`uni-program/student.h`

```
#ifndef STUDENT_H
#define STUDENT_H

#include <string>
#include <string_view>
#include "uniperson.h"

class Student : public UniPerson  // non inheritance from UniPerson
{
private:
    double tuition;
    // We can remove a lot of code here!!!
public:
    Student(std::string_view name = "", int age = 0, double tuition = 42000);
    double get_tuition() const;
};

#endif
```

Listing 12 Implementation file for class `Student` Filename=`uni-program/student.cpp`

```cpp
#include "student.h"

Student::Student(std::string_view name, int age, double tuition)
    : UniPerson{name, age}, tuition{tuition}
    {}

double Student::get_tuition() const {
    return tuition;
}
```

Listing 13 Header file for class `UGStudent` Filename=`uni-program/ugstudent.h`

```cpp
#ifndef UGSTUDENT_H
#define UGSTUDENT_H

#include "student.h"

class UGStudent : public Student {
public:
    // Call the base Student constructor
    UGStudent(std::string_view name = "", int age = 0, double tuition = 0.0);

    void write_BSc_thesis() const;
};

#endif
```

Listing 14 Implementation file for class `UGStudent` Filename=`uni-program/ugstudent.cpp`

```cpp
#include <iostream>
#include "ugstudent.h"

// Remember: No default arguments here!
UGStudent::UGStudent(std::string_view name, int age, double tuition)
    : Student{name, age, tuition} {}

void UGStudent::write_BSc_thesis() const {
    std::cout << "Student (ID: " << get_id() << ") is writing the thesis" << std::endl;
}
```

Listing 15 Two simple classes `Base` and `Derived` which inherits from `Base`. Filename=`base_and_derived.cpp`

```cpp
class Base
{
public:
    int id {};

    Base(int id=0)
        : id { id }
    {
    }

    int get_id() const { return id; }
};

class Derived: public Base
{
public:
    double value {};

    Derived(double value=0.0)
        : value { value }
    {

    }

    double get_value() const { return value; }
};

int main()
{
    Base base{ 42 };
    Derived derived { 42.24 };
    return 0;
}
```

Listing 16 The constructor of the `Base` class is executed before the constructor of the `Derived` class. Filename=`parent_and_child_with_main.cpp`

```cpp
#include <iostream>
class Base
{
public:
    int id {};

    Base(int id=0)
        : id { id }
    { std::cout << "Base object is constructed!\n"; }

    int get_id() const { return id; }
};

class Derived: public Base
{
public:
    double value {};

    Derived(double value=0.0)
        : value { value }
    { std::cout << "Derived object is constructed!\n"; }

    double get_value() const { return value; }
};

int main ()
{
    std::cout << "Instantiating a Parent object!\n";

    Base parent;
    std::cout << "==============================\n";
    std::cout << "Instantiating a Derived object!\n";
    Derived child;

    return 0;
}
```

Listing 17 Order of construction of derived classes in an inheritance chain. Filename=`inheritance_chain_four_classes.cpp`

```cpp
#include <iostream>

class A
{
public:
    A() { std::cout << "\tConstructor of A\n"; }
};

class B: public A
{
public:
    B() { std::cout << "\tConstructor of B\n"; }
};

class C: public B
{
public:
    C() { std::cout << "\tConstructor of C\n"; }
};

class D: public C
{
public:
    D() { std::cout << "\tConstructor of D\n"; }
};

int main()
{
    std::cout << "Constructing A: \n";
    A a;
    std::cout << "Constructing B: \n";
    B b;
    std::cout << "Constructing C: \n";
    C c;
    std::cout << "Constructing D: \n";
    D d;
}
```

Listing 18 An example illustrating the initialization process of a derived class. Filename=`initialization_of_derived_class.cpp`

```cpp
class Base
{
public:
    int id {};

    Base(int id=0) : id{ id }  {}

    int get_id() const { return id; }
};

class Derived: public Base
{
public:
    double value {};

    Derived(double value=0.0) : value{ value } {}

    double get_value() const { return value; }

};

int main()
{
    // With non-derived classes, constructors only have to worry about
    // their own members.
    Base parent { 42 };   // use Base(int) constructor

    // How about the the "id" member that child inherits from Base?
    Derived child { 42.24 };  // use Derived(double) constructor

    return 0;
}
```

Listing 19 Call Base constructor in the Derived class constructor. Filename=`initialization_of_derived_class_base_constructor.cpp`

```cpp
#include <iostream>
class Base
{
public:
    const int id {};

    Base(int id=0) : id{ id }  {}

    int get_id() const { return id; }
    void print() const {
        std::cout << "Base:\n\tId: " << id;
    }
};

class Derived: public Base
{
public:
    double value {};

    Derived(double value=0.0, int id=0)
        : Base{id}, value{ value } {}

    double get_value() const { return value; }

    void print() const {    // this will override void Base::print()
        std::cout << "Derived:\n\tId: " << id << '\n';
        std::cout << "\tValue: " << value;
    }
};

int main()
{
    Base base{ 4224 };
    base.print();
    Derived derived{ 42, 24 };
    derived.print();
    std::cout << '\n';
    return 0;
}
```

Listing 20 Calling base constructor in member initializer list of derived classes can be done in inheritance chain. Filename=`initializing_base_members_inheritance_chain.cpp`

```cpp
#include <iostream>
class A
{
public:
    A(int a) {
        std::cout << "A: " << a << '\n';
    }
};

class B: public A
{
public:
    B(int a, double b) : A{ a } {
        std::cout << "B: " << b << '\n';
    }
};

class C: public B
{
public:
    C(int a, double b, char c) : B{ a, b } {
        std::cout << "C: " << c << '\n';
    }
};

int main()
{
    C c{ 5, 4.3, 'R' };
    return 0;
}
```

Listing 21 Calling base constructor in member initializer list of derived classes can be done in inheritance chain. Filename=`initializing_base_members_inheritance_chain.cpp`

```cpp
#include <iostream>
class A
{
public:
    A(int a) {
        std::cout << "A: " << a << '\n';
    }
};

class B: public A
{
public:
    B(int a, double b) : A{ a } {
        std::cout << "B: " << b << '\n';
    }
};

class C: public B
{
public:
    C(int a, double b, char c) : B{ a, b } {
        std::cout << "C: " << c << '\n';
    }
};

int main()
{
    C c{ 5, 4.3, 'R' };
    return 0;
}
```

Listing 22 Besides inheriting members from the base class, the derived class can add more functionalities. Filename=`add_functionality_example.cpp`

```cpp
#include <iostream>

class Base
{
protected:
    int id {};

public:
    Base() = default; // we need default constructor here, see Derived class

    Base(int id): id { id }{}

    void identify() const { std::cout << "I am a Base. ID = " << id << '\n'; }
};

class Derived : public Base
{
    double value;
public:
    Derived(double value) : value {value} {}    // no id argument
    Derived(double value, int id) : Base{id}, value {value} {}

    double get_value() const { return value; }
    void print() const {
        std::cout << "ID: " << id << ", value: " << value;
    }
};

int main()

{
    Derived d1 { 42.0 };
    Derived d2 { 24.0, 42 };
    d1.identify();
    d2.identify();
    d1.print(); std::cout << '\n';
    d2.print(); std::cout << '\n';

    return 0;
}
```

Listing 23 Same as before but a slightly more complex. Filename=`add_functionality_example_vector.cpp`

```cpp
#include <iostream>
#include <vector>

class Base
{

protected:
    int id {};

public:

    Base() = default; // we need default constructor here, see Derived class

    Base(int id): id { id }{}

    void identify() const { std::cout << "I am a Base. ID = " << id << '\n'; }
};

class Derived : public Base

{
    double value {};

public:
    Derived() = default;

    Derived(double value) : value {value} {}    // no id argument
    Derived(double value, int id) : Base{id}, value {value} {}

    double get_value() const { return value; }
    void print() const {

        std::cout << "ID: " << id << ", value: " << value;
    }

};

int main()
{

    std::vector<Derived> list;
    list.push_back(Derived {42.0, 24});

    list.push_back(Derived {24.0, 42});
    list.push_back(Derived {});

    for (auto element : list) {

        element.print();
        std::cout << '\n';

    }
```

Listing 24 Derived classes can call their base class functions. Filename=`call_base_class_function.cpp`

```cpp
#include <iostream>

class Base
{
public:
    Base() { }

    void identify() const { std::cout << "Base::identify()\n"; }
};

class Derived: public Base
{
public:
    Derived() { }
};

int main()
{
    Base base {};
    base.identify();

    Derived derived {};
    derived.identify();

    return 0;
}
```

Listing 25 The derived class can override the member function of the base class. It can redefine behaviors of the base class. Filename=`call_base_class_function.cpp`

```cpp
#include <iostream>

class Base
{
public:
    Base() { }

    void identify() const { std::cout << "Base::identify()\n"; }
};

class Derived: public Base
{
public:
    Derived() { }

    void identify() const { std::cout << "Derived::identify()\n"; }
};

int main()
{
    Base base {};
    base.identify();

    Derived derived {};
    derived.identify();

    return 0;
}
```

Listing 26 We can call the base class's member function and add more to it. Filename=`add_to_existing_functionality.cpp`

```cpp
#include <iostream>

class Base
{
public:
    Base() { }

    void identify() const { std::cout << "Base::identify()\n"; }
};

class Derived: public Base
{
public:
    Derived() { }

    void identify() const
    {
        std::cout << "Derived::identify()\n";
        Base::identify(); // note call to Base::identify() here
    }
};

int main()
{

    Base base {};
    base.identify();

    Derived derived {};
    derived.identify();

    return 0;
}
```

Listing 27 With `using` declaration we can change the access level of derived class' members. Filename=`change_inherited_member_access_level.cpp`

```cpp
#include <iostream>

class Base
{
private:

    int id {};
public:
    Base(int id) : id {id} {}
protected:
    void print_id() const { std::cout << "ID: " << id; }

};

class Derived: public Base
{
private:

    double value {};

public: // note this public, everything below this is public now
    Derived(double value, int id) : Base {id}, value {value} {}
    double get_value() const { return value; }

    //Base::print_id() was inherited as protected, so the public has no access.
    // But we are changing it to public via a using declaration
    using Base::print_id;   // note: no parenthesis here
};


int main()
{
    Derived d {24.42, 42};
    d.print_id();      // prints 7
    std::cout << ", d.value: " << d.get_value() << '\n';

}
```

Listing 28 With `using` declaration we can also high functionality that exists in the base class. Filename=`hide_functionality.cpp`

```cpp
#include <iostream>
class Base
{
public:
    int id {};
public:
    double public_mem {};   // this is a public member

    Base(int id) : id {id} {}
protected:
    void print_id() const { std::cout << "ID: " << id; }
};

class Derived: public Base
{
private:
    double value {};
    using Base::public_mem; // this is now private in Derived.

public: // note this public, everything below this is public now
    Derived(double value, int id) : Base {id}, value {value} {}
    double get_value() const { return value; }
    using Base::print_id;   // this is now public to the outside world
};

int main()
{   Base b { 42 };
    std::cout << "b.public_mem: " << b.public_mem << '\n'; // This is OK
    Derived d {24.42, 42};
    // std::cout << d.public_mem;      // ERROR: public_mem is private in Derived

}
```

Listing 29 We can `delete` a function inherited from the base class in the derived class. This function cannot be called at all through a derived object. Filename=`delete_functions_in_derived.cpp`

```cpp
#include <iostream>
class Base
{
private:
    int id {};

public:
    Base(int id) : id { id } { }
    int get_id() const { return id; }
    void print_base() const {
        std::cout << "Base object with id: " << id;
    }
};

class Derived : public Base
{
    double value {};
public:
    Derived(double value, int id) : Base {id}, value {value} {}
    double get_value() const { return value; };

    // it does not make sense to print a Derived-type object
    // using the print_base() for a Base-type object
    void print_base() const = delete; // mark this function as inaccessible
};

int main()
{
    Derived derived { 42.24, 42 };
    std::cout << derived.get_id() << '\n';  // this is fine.

    // ERROR: This function has been deleted in Derived
    // std::cout << derived.print_base();

    return 0;
}
```

Listing 30 `SmartPhone` inherits both members from `Camera` and `Phone`. Filename=`multiple_inheritance_smartphone.cpp`

```cpp
#include <iostream>

class Camera {
public:
    void take_photo() {
        std::cout << "Taking a photo!" << std::endl;
    }
};

class Phone {
public:
    void make_call() {
        std::cout << "Making a phone call!" << std::endl;
    }
};

// Multiple Inheritance
class SmartPhone : public Camera, public Phone {
    // Inherits members from both Camera and Phone

    // add more functionality to phone
    void use_app(std::string app_name) {
        std::cout << app_name << "is being used.\n";
    }
};

int main() {
    SmartPhone my_phone;

    my_phone.take_photo();
    my_phone.make_call();

    return 0;
}
```